HALOPERIDOL

Extrapyramidal effects
QT prolongation

CHEMICAL NAME 4-[4-(4-chlorophenyl)-4-hydroxypiperidin-1-yl]-1-(4-fluorophenyl)butan-1-one
CLASS First-generation (typical) butyrophenone antipsychotic
TRADE NAMES Haldol · Serenace · Haldol Decanoate (depot)

THERAPEUTIC USES / INDICATIONS

- Schizophrenia and acute psychosis; mania.
- Acute agitation / delirium (short-term).
- Nausea and vomiting (palliative, PONV); Tourette's syndrome, intractable hiccups.
- Depot for maintenance in schizophrenia.

ADMINISTRATION

- **PO** tablets/liquid; **IM** (lactate) for acute agitation.
- **IV** (off-label in some countries) — ECG monitoring.
- **Decanoate depot:** deep IM every 4 weeks — never IV.
- **SC** infusion in palliative care.

DOSAGES

- **Acute agitation:** 2–5 mg IM, repeat hourly if needed (max 20 mg/day; elderly 0.5–1 mg).
- **Delirium (elderly):** 0.5–1 mg PO/IM.
- **Schizophrenia:** 2–10 mg/day PO.
- **Nausea (palliative):** 0.5–1.5 mg at night.
- **Depot:** 50–300 mg every 4 weeks.

PHARMACOKINETICS

- **Onset:** IM 20–30 min; PO 2–6 h. **Duration:** 24 h.
- Metabolised by CYP3A4, 2D6.
- **t½:** 15–37 h (depot ~3 weeks).

PHARMACODYNAMICS

- Potent **dopamine D2 antagonist** in mesolimbic pathway (antipsychotic) and CTZ (antiemetic).
- D2 block in nigrostriatal pathway → EPS; tuberoinfundibular → ↑ prolactin.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Extrapyramidal symptoms: acute dystonia, akathisia, parkinsonism.
- Sedation (less than chlorpromazine).
- Hyperprolactinaemia.

ADVERSE EFFECTS (SERIOUS)

- QT prolongation, torsades, sudden death (IV, high doses).
- Neuroleptic malignant syndrome: fever, rigidity, altered mental state, ↑ CK.
- Tardive dyskinesia.
- ↑ stroke and mortality in elderly with dementia.

CONTRAINDICATIONS & CAUTIONS

- Parkinson's disease, Lewy body dementia (severe sensitivity).
- QT prolongation, recent MI, uncompensated heart failure.
- Comatose states, CNS depression.
- Basal ganglia lesions.

KEY INTERACTIONS

- **QT-prolonging drugs:** avoid.
- **Levodopa:** antagonism.
- **Carbamazepine, rifampicin:** ↓ levels; **CYP inhibitors:** ↑ levels.
- **Lithium:** neurotoxicity (rare).

NURSING MANAGEMENT

- Baseline ECG (QTc) and electrolytes when possible.
- Monitor for dystonia (treat with procyclidine), akathisia, rigidity, temperature.
- Monitor sedation, BP, falls.
- Use non-drug delirium measures first.

PATIENT EDUCATION

- Report muscle spasms, restlessness, stiffness, fever.
- Avoid alcohol; may cause drowsiness.
- Depot: keep appointments.

KEY POINTS

- Potent D2 antagonist (typical).
- EPS, NMS, QT prolongation.
- Agitation 2–5 mg IM (0.5–1 mg elderly).
- Avoid in Parkinson's/Lewy body.